POLYTWELVE

Programmable 12-voice synthesizer · User Manual · version 0.8.0



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1. Introduction

Polytwelve is a programmable 12-voice analog-style synthesizer by Ticco Ross (2026), version 0.8.0. Each voice has one oscillator with a sub-oscillator, a resonant 24 dB/oct low-pass filter, an envelope, an amplifier and its own modulation generator (MG). The twelve voices are summed and pass through a stereo effects section (Chorus, Phaser or Ensemble).



Signal flow: one voice of twelve, then the shared effects. The EG shapes the filter and the amplifier; the MG (dashed) reaches one destination at a time, chosen by its MODE switch.

It plays three ways: Poly (twelve notes at once), Unison (all twelve voices on one note) and Chord (one key plays a stored chord). A built-in arpeggiator, a preset browser with favourites and per-voice analog drift complete it.

Formats

System	Formats	Installer
macOS 10.13 or later (Apple Silicon and Intel)	Audio Unit, VST3, standalone app	Polytwelve-0.8.0-macOS.pkg
Windows 10 or later (64-bit)	VST3, standalone app	Polytwelve-0.8.0-Windows-Setup.exe
Linux x86_64	VST3, LV2, standalone app	Polytwelve-0.8.0-Linux-x86_64.tar.gz

2. Installing

macOS security warning. The macOS installer is not notarised by Apple, so the first time you open it macOS says it “cannot be opened because Apple cannot check it for malicious software” (or that it is from an unidentified developer). This is expected. To install it: click **Done** (not Move to Trash), open **System Settings → Privacy & Security**, scroll down to the message about Polytwelve and click **Open Anyway**, then confirm with your password. On older macOS you can instead right-click (or Control-click) the .pkg and choose **Open**.

- **macOS:** double-click the .pkg and follow the steps. Click Customize to skip the AU, VST3 or app.
- **Windows:** run the Setup .exe. If SmartScreen appears, click **More info → Run anyway**. The VST3 goes to *C:\Program Files\Common Files\VST3*.
- **Linux:** unpack the archive and run *./install.sh*. It installs for your user to *~/.vst3*, *~/.lv2* and *~/.local/bin*, with no root needed.

Afterwards, rescan plug-ins in your DAW. In Ableton Live on a Mac: Settings → Plug-Ins, turn on Use Audio Units v2, hold Option and click Rescan. Polytwelve appears under Ticco Ross.

3. The panel

The panel follows the sound from left to right: the preset strip and logo along the top, the sound sections in the middle row, and the playing sections in the bottom row. The window can be resized by dragging its bottom-right corner; it keeps its shape.

Row	Sections, left to right
Top	PRESET strip (display, < > SAVE BANK ★) · POLYTWELVE logo (click it for About)
Middle	VCO · VCF · VCA · EG · MG
Bottom	WHEELS · VOICE · ARPEGGIATOR · EFFECTS · MASTER

Working the controls

- **Knobs:** drag up or down (or left and right). While you drag, a bubble shows the exact value (Hz, ms, dB, semitones). Double-click returns a knob to its default.
- **Slide switches:** click a position name, or click the switch and drag; the mouse wheel moves it too.
- **Rubber buttons:** click to turn on or off. The red LED above shows it is on; the EFFECTS buttons also light up orange.
- **LED displays** (SYNC RATE, BEND RANGE): click the < > buttons beside them, drag over the display, or scroll over it.
- Every control can be automated from your DAW and is saved with your project and with presets.

4. VCO, VCF and VCA

VCO (oscillator)

Each voice has one oscillator plus a sub-oscillator. The waveforms are band-limited, so high notes stay clean.

Control	Range	What it does
PW/PWM	0–10	In Pulse mode, the pulse width: 0 = square (50%), 10 = thin (95%). In PWM mode, how far the width sweeps. No effect on Saw.
PWM SPEED	0.1–12.8 Hz	How fast the width sweeps in PWM mode.
OCTAVE	16' · 8' · 4'	Shifts the oscillator down one octave, normal, or up one octave.
WAVEFORM	SAW · PULSE · PWM	Saw: bright and full. Pulse: hollow; set its width with PW/PWM. PWM: a pulse whose width sweeps on its own, for a moving, chorused tone.
SUB OSC	OFF · 1 OCT · 2 OCT	Adds a square wave one or two octaves below the main oscillator, at equal level.

Tip: Pulse with the sub at 1 OCT and a low cutoff gives classic Polysix basses. PWM with Ensemble gives lush strings.

VCF (filter)

A 4-pole (24 dB/oct) resonant low-pass filter with soft saturation in its feedback loop. Resonance stays in tune across the range, and at high settings the filter self-oscillates into a clean sine.

Control	Range	What it does
CUTOFF	27.5 Hz – 17 kHz	The filter's brightness. Lower = darker.
RESONANCE	0–10	Emphasis at the cutoff. Near 10 the filter sings on its own.
EG INTENSITY	−5 to +5 on the scale	How far the envelope moves the cutoff: up to ±5 octaves. Negative values sweep it down instead of up.
KBD TRACK	0–10	How much the cutoff follows the keys. At 10 it tracks the keyboard one-to-one, centred on C3.

VCA (amplifier)

Control	Range	What it does
MODE	EG · GATE	EG: volume follows the envelope. GATE: volume is fully on while a key is held and off when released, organ-style (the EG still drives the filter).
ATTENUATOR	±10 dB (−5 to +5 on the scale)	Level of each voice before the effects. Lower it if high resonance or Unison gets too loud.

5. EG and MG

EG (envelope)

One ADSR per voice, shared by the filter (through EG INTENSITY) and the amplifier. Its curves are exponential, like an analog envelope.

Control	Range	What it does
ATTACK	1 ms – 10 s	Time to rise to full level.
DECAY	1 ms – 10 s	Time to fall to the sustain level.
SUSTAIN	0–10	Level held while the key is down.
RELEASE	1 ms – 10 s	Time to fade after the key is released.

The time knobs are logarithmic: 2.5 ≈ 10 ms, 5 ≈ 100 ms, 7.5 ≈ 1 s, 10 = 10 s. With ANALOG turned up, each voice's decay and release differ slightly.

MG (modulation generator)

A triangle LFO, one per voice, that fades in after each note starts and modulates one destination. With ANALOG turned up, each voice's MG runs at a slightly different rate.

Control	Range	What it does
FREQUENCY	0.01–20 Hz	Speed.
DELAY	1 ms – 3.2 s	How long the modulation takes to fade in after a note starts. Restarts with every note.
LEVEL	0–10	Depth. The curve is gentle at the bottom for fine vibrato.
MODE	VCO · VCF · VCA	VCO: vibrato (up to ±12 semitones at full LEVEL). VCF: filter wobble (up to ±4 octaves). VCA: tremolo.

The MOD wheel always adds vibrato from the same triangle, up to ±2 semitones, whatever the MODE, LEVEL or DELAY settings.

6. Voice and Chord Memory

Control	Range	What it does
MODE	POLY · UNISON · CHORD	How keys are assigned to voices (below).
STORED CHORD	display	The chord in memory: its name on top (e.g. Am7, Cmaj7, C/E for an inversion) and its notes below, lowest first. Bright in Chord mode, dim otherwise.
ANALOG	0–10	Per-voice variation: a small fixed tuning offset and slow pitch drift, plus differences in cutoff, decay, release and MG and PWM rates. 0 = perfectly stable.
DETUNE	0–10	A fixed tuning offset per voice, up to ±25 cents. Most audible in Unison.
GLIDE	Off, 5 ms – 3 s	Portamento: each new note slides in pitch from the previous one, taking about this long. 0 = off.
HOLD	on / off	Latches notes: they keep sounding after you let go. A sustain pedal does the same.
LEGATO	on / off	Notes glide only if you press the new key before releasing the old one.
MEMORY	button	Stores the chord you are holding and switches to Chord mode.

Voice modes

- **POLY:** up to twelve notes at once. Each new note takes the next free voice in turn, which lets release tails ring. With all twelve busy, the oldest released voice is reused first, then the oldest held one.
- **UNISON:** all twelve voices play one note, stacked with their DETUNE and ANALOG offsets, for a huge mono sound. The last key pressed wins; releasing it returns to a key still held.
- **CHORD:** every key plays the stored chord, transposed. One chord sounds at a time, with the same last-key behaviour as Unison.

Storing a chord

1. Hold the chord on your MIDI keyboard, e.g. A2, C3, E3, G3.
2. While holding it, press **MEMORY**. Its LED blinks.
3. Let go. MODE is now CHORD, the MEMORY LED stays lit and the STORED CHORD display shows Am7. Pressing D3 plays Dm7, F3 plays Fm7, and so on.

- MEMORY with no keys held switches to Chord mode using the chord already stored.
- Moving MODE to CHORD while holding two or more keys also stores them.
- The chord is saved with presets and with your project. The default chord is a minor triad (C3 · Eb3 · G3).
- Note names follow Ableton's convention (middle C = C3).

7. Arpeggiator

The arpeggiator plays the keys you hold one at a time, in pitch order, repeated over a range of octaves. Each step sounds for half its length.

Control	Range	What it does
ARP	on / off	Turns the arpeggiator on.
LATCH	on / off	Keeps the pattern running after you let go. The next chord you play replaces it.
SYNC	on / off	Follows your DAW's tempo at the SYNC RATE instead of the SPEED knob.
SPEED	0.5–24 steps per second	Rate when SYNC is off.
RANGE	1 OCT · 2 OCT · FULL	How many octaves the pattern covers: one, two, or four.
MODE	UP · DOWN · UP/DN	Direction. UP/DN doesn't repeat the top and bottom notes.
SYNC RATE	1/4 · 1/8 · 1/8T · 1/16 · 1/16T · 1/32	Step length when SYNC is on. The display dims while SYNC is off.

With SYNC on and the DAW playing, steps lock to the song's beat grid. With the transport stopped, it runs at the project tempo. In Unison each step is the full twelve-voice stack, and in Chord each step plays the stored chord.

8. Effects, Wheels and Master

Effects

One effect at a time processes all twelve voices together and makes the output stereo. Press OFF, CHORUS, PHASER or ENSEMBLE; the selected button lights up. SPEED/INT sets its speed or intensity.

Effect	SPEED/INT controls	Sound
OFF	—	Dry, mono in the centre.
CHORUS	Speed, 0.1–4 Hz	A modulated delay, opposite on left and right. Widens leads and pads.
PHASER	Speed, 0.05–6.4 Hz	A six-stage phaser with feedback, its sweep offset between left and right.
ENSEMBLE	Intensity	Three modulated delay lines with slow and fast movement, for the lush string-machine sound.

Wheels

Control	Range	What it does
PITCH	up / down	Drag to bend all voices. Springs back to the centre when you let go.
MOD	0–max	Drag up to add MG vibrato, up to ±2 semitones. Stays where you leave it.

The cream mark on each wheel shows its position. The wheels also move when your MIDI keyboard sends pitch bend or mod wheel.

Master

Control	Range	What it does
VOLUME	0–10 (default 7.5)	Overall output level, after the effects.
TUNE	±100 cents (−5 to +5 on the scale)	Fine-tunes all voices by up to one semitone either way.

Control	Range	What it does
BEND RANGE	0–12 semitones (default 2)	How far the PITCH wheel and MIDI pitch bend reach, up and down.
MODEL	CLASSIC · POLYSIX	Which sound engine plays: see the next chapter.

Sound engine: MODEL

Polytwelve has two sound engines, picked with the MODEL switch in the MASTER section. Every preset plays on both, so you can flip the switch and compare. MODEL belongs to the instrument, not to a sound: changing preset keeps it, and your project remembers it.

- **CLASSIC** is the original Polytwelve engine, ported from the Polysix 2 Reaktor ensemble, with its ladder filter and stereo effects. Out of the box it sounds as in earlier versions; the CLASSIC Calibration switches below let you move it, one part at a time, towards a real Polysix.
- **POLYSIX** is built from the Korg Polysix service manual (circuit diagrams KLM-366 and KLM-368, and the factory adjustment procedure).

What POLYSIX changes

Part	POLYSIX model
Filter	An SSM2044 model, the chip on the Polysix voice board: four transconductance stages that each saturate softly, and a saturating resonance loop. At RESONANCE 10 it settles into a clean sine. The bass thins as resonance rises, as on the original, so very resonant sounds play quieter.
Cutoff	The knob follows the curve measured on a real Polysix: open from 7 up, much darker at 5, almost shut at 4 and closed at 2. KBD TRACK pivots low on the keyboard, as on the real unit.
Pulse / PWM	PW narrows all the way until the pulse closes at 10; PWM sweeps through that point, about 5 Hz at SPEED 5. Pulse and PWM levels as measured.
Sub	Sub oscillator level as measured (about 3 dB under the old level).
MG	FREQUENCY about 0.6 Hz at 2.5, 3.2 Hz at 5, 16 Hz at 7.5; depth a third of the old one; DELAY reaches half depth after about 1.6 s at 5.
Envelopes	Time knobs steeper: very short at the bottom, ATTACK 10 about 10 s, RELEASE 10 about 19 s.
Effects	Modelled on the KLM-368 effect board: bucket-brigade (MN3004) delay lines with their filters, soft compander saturation and a little noise; mono, like the Polysix output. CHORUS: one delay, 2–4 ms. PHASE: a short swept delay (0.5–2.6 ms) with feedback, as on the original, which is a flanging sound rather than a phaser. ENSEMBLE: three delays driven by a slow and a fast modulation, 120° apart.

Since 0.8.0 the POLYSIX model is fitted to recordings of a real Polysix (27 test takes: levels, filter curve, resonance, envelopes, MG and all three effects).

CLASSIC Calibration (A/B)

Open the preset menu (BANK) and choose **CLASSIC Calibration (A/B)**. Each item switches one fix, measured on a real Polysix, into the CLASSIC engine, so you can compare it and keep only the ones you like. All are off by default. They are global like MODEL: presets don't change them, projects remember them. They have no effect on POLYSIX, which always uses the first seven.

Switch	What it changes
Cutoff Curve	The real CUTOFF knob law and KBD TRACK pivot.

Switch	What it changes
Resonance Level	Self-oscillation about 8 dB louder, as on the real unit.
Pulse Width Range	Pulse closes at PW 10; real pulse/PWM levels; PWM SPEED range.
Sub Level	Sub oscillator level as measured.
MG Rate & Depth	Real FREQUENCY curve, a third of the depth, longer DELAY.
Envelope Times	Steeper attack, decay and release time curves.
Effects	Uses the fitted bucket-brigade effects (mono) with the real speeds and levels.
Noise Floor	Adds the real unit's faint hiss and slow ripple (about −56 dB). Never on in POLYSIX, and “All On” leaves it off: switch it on by itself.

9. Presets

A preset stores every control plus the stored chord. The amber display shows the bank (FACTORY or USER), the preset's number and its name.

Control	What it does
Display	Click it to open the preset browser. Scroll or drag over it to step through presets.
< >	Previous / next preset.
SAVE	Saves the current sound as a user preset. It asks for a name, and asks before replacing an existing one.
BANK	Opens the preset browser: Favourites, Factory and User lists, plus Save As..., Delete, Step Through Favourites Only, Open Preset File... and Show Preset Folder.
★	Adds or removes the current preset from favourites. It glows when the preset is a favourite.

With **Step Through Favourites Only** ticked, < and > step through favourites only, and the display shows ★ ONLY. Your project remembers which preset was loaded. Factory presets also appear as programs in hosts that show a program list.

Factory presets

Preset	Character
Init	Plain saw, the starting point
Strings	Slow saw pad through the Ensemble
Brass	Swelling filter envelope
Bass	16' pulse with sub, resonant pluck
Square Lead	Unison square with legato glide, delayed vibrato and chorus
Rezzy Arp	Resonant saw on a synced 1/16 arpeggio, through the phaser
Filter Sweep	PWM with the MG slowly sweeping the filter
Minor Chord	Chord mode with a minor triad, plucky, with chorus

Preset files

User presets are .p6preset files, one per sound, which you can copy, back up or share. Favourites are listed in favourites.txt in the same folder.

System	Preset folder
macOS	~/Library/Audio/Presets/Ticco Ross/Polytwelve
Windows	Documents\Ticco Ross\Polytwelve\Presets
Linux	~/.local/share/Ticco Ross/Polytwelve/Presets

10. MIDI implementation

Polytwelve responds on all MIDI channels.

Message	Effect
Note on / off	Plays notes. Velocity is ignored, as on the original Polysix.
Pitch bend	Bends all voices by up to the BEND RANGE; the on-screen PITCH wheel follows it.

Message	Effect
CC 1 (mod wheel)	Adds MG vibrato, up to ±2 semitones; the on-screen MOD wheel follows it.
CC 64 (sustain pedal)	Works like HOLD.
CC 120 / CC 123	All sound off / all notes off.

All 37 controls are exposed to the host for automation and MIDI mapping. In Ableton Live, use the plug-in's Configure button to add them to the device, then map them like any Live control.

11. Specifications, uninstalling and contact

Item	Value
Version	0.8.0 (shown under the PROGRAMMABLE 12 VOICE SYNTHESIZER line next to the logo, and in About)
Polyphony	12 voices
Per voice	1 oscillator (saw, pulse, PWM) + sub-oscillator, 24 dB/oct resonant low-pass filter, ADSR, VCA, triangle MG, glide
Effects	Chorus, Phaser, Ensemble (stereo)
Formats	AU (macOS), VST3, LV2 (Linux), standalone app
Parameters	37, all automatable

Uninstalling

- **macOS:** delete Polytwelve.component from /Library/Audio/Plug-Ins/Components, Polytwelve.vst3 from /Library/Audio/Plug-Ins/VST3 and Polytwelve.app from Applications.
- **Windows:** Settings → Apps → Polytwelve → Uninstall.
- **Linux:** run ./uninstall.sh from the downloaded folder.

Uninstalling leaves your presets in place.

Contact

Polytwelve is designed by Ticco Ross, 2026. Questions, bug reports and suggestions: **ticcoross@gmail.com**. More instruments: **ticcoross.gumroad.com**. Click the POLYTWELVE logo on the panel for the About window.